

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	AIDS SLOT-B	Date:	15-08-2026

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.

A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.

Ans:

Query	Semantic Representation	Action	Object
Q1	ACTIVATE (Roaming, Customer)	Activate	Roaming
Q2	DEACTIVATE (Caller Tune, Customer)	Deactivate	Caller Tune
Q3	QUERY (Data Balance, Customer)	Query	Data Balance
Q4	ACTIVATE (5GService, Customer)	Activate	5G Service

The semantic representation connects the customer as the entity with the required action and service/object.

2. Determine which query contains semantic interpretation errors.

Ans:

Q2 contains the error.

- Actual intent: Deactivate Caller Tune
- Predicted intent: Activate Caller Tune

The action is incorrectly interpreted as ACTIVATE instead of DEACTIVATE.

3. Evaluate how meaning representation affects chatbot decision accuracy.

Ans:

Correct semantic representation helps the chatbot identify the user's exact action and object. An incorrect representation can cause the chatbot to perform the wrong operation.

For example:

DEACTIVATE (Caller Tune, Customer) → Caller Tune is stopped.

But:

ACTIVATE (Caller Tune, Customer) → Caller Tune is enabled.

Therefore, accurate meaning representation directly improves chatbot decision accuracy.

4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

Ans:

- Use intent classification with semantic features.
- Apply contextual language understanding.
- Identify action words such as activate, deactivate, check, cancel.
- Use entity recognition for services such as 5G, roaming, caller tune.
- Use context from previous conversations.
- Validate the predicted action before executing it.

Conclusion: Semantic representation allows the chatbot to convert natural-language requests into structured actions. Accurate action and entity identification is essential for correct telecom service operations.

2. First-Order Predicate Calculus for Smart Manufacturing.

An Industry 4.0 manufacturing plant monitors machines using semantic rules. Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

Tasks:

1. Represent the production data using First-Order Predicate Calculus.

Ans:

Given:

- $\text{Active}(\text{M1})$
- $\text{Active}(\text{M2})$
- $\text{Maintenance}(\text{M3})$
- $\text{Active}(\text{M4})$

Rules:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

From the data:

- $\text{Active}(\text{M1}) \rightarrow \text{Producing}(\text{M1})$
- $\text{Active}(\text{M2}) \rightarrow \text{Producing}(\text{M2})$
- $\text{Maintenance}(\text{M3}) \rightarrow \neg \text{Producing}(\text{M3})$
- $\text{Active}(\text{M4}) \rightarrow \text{Producing}(\text{M4})$

2. Analyse which products are currently available.

Ans:

The rule for availability is: **$\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$**

Therefore, a product is available only when its producing machine is active.

For example: **$\text{Produces}(\text{M1}, \text{Gear}) \wedge \text{Active}(\text{M1}) \rightarrow \text{Available}(\text{Gear})$**

Thus, products produced by **M1, M2 and M4** can be considered available, provided their production relationship is given.

3. Infer whether Gear production is affected by maintenance activities.

Ans:

If Gear is produced by M3: **Maintenance(M3) $\rightarrow$ $\neg$ Producing(M3)**

Therefore: **$\neg$ Producing(M3) $\rightarrow$ Gear production is affected.**

Hence, if M3 is responsible for Gear production, Gear production will stop during maintenance.

4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

Ans:

Predicate logic is useful because it:

- Represents machine states clearly.
- Expresses relationships between machines and products.
- Supports logical inference.
- Detects production problems.
- Helps automated decision-making.

Conclusion: First-Order Predicate Calculus provides a structured way to represent machine states and infer production conditions. It can support real-time industrial decision-making.

3. Word Sense Disambiguation in E-Commerce Search Engines. An e-commerce company observed the following search logs.

"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyse the contextual information and determine the correct word sense for each query.

Ans:

Query	Correct Sense	Evidence
Apple accessories	Technology Brand	iPhone Charger
Mouse wireless	Computer Device	Bluetooth Mouse
Java tutorial	Programming Language	Coding Lessons
Python course	Programming Language	Software Development Training

The clicked results provide strong contextual evidence for selecting the correct sense.

2. Identify the semantic cues used for disambiguation.

Ans:

The main cues are:

- **Apple** $\rightarrow$ **iPhone Charger** $\rightarrow$ technology/brand
- **Mouse** $\rightarrow$ **Bluetooth Mouse** $\rightarrow$ computer device
- **Java** $\rightarrow$ **Coding Lessons** $\rightarrow$ programming language
- **Python** $\rightarrow$ **Software Development Training** $\rightarrow$ programming language

Other useful cues include surrounding words such as accessories, wireless, tutorial, and course.

3. Evaluate the impact of incorrect sense selection on search engine performance.

Ans:

Incorrect word sense selection can:

- Return irrelevant search results.
- Reduce search accuracy.
- Decrease user satisfaction.
- Reduce click-through rate.
- Produce incorrect recommendations.
- Affect e-commerce sales.

For example, interpreting Python course as a snake-related query would return irrelevant results instead of programming courses.

4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

Ans:

A large e-commerce system can use:

User Query → Preprocessing → Context Analysis → Candidate Senses → WSD Model → Correct Sense → Search/Recommendation

The system can combine:

- Contextual embeddings.
- User search history.
- Click behaviour.
- Product metadata.
- Query frequency.
- Machine-learning-based WSD.
- Feedback from user interactions.

Conclusion: WSD improves search relevance by selecting the correct meaning of ambiguous words. Combining linguistic context with user and product information can improve recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems. A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient

Headache	Symptom
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Tasks:

1. Analyse how syntactic structures contribute to semantic role assignment.

Ans:

Syntactic structure identifies the relationships between words.

Example: Doctor prescribed medicine to patient.

- Subject → Doctor
- Verb → prescribed
- Object → medicine
- Recipient → patient

Therefore:

Doctor → Agent

Medicine → Object/Theme

Patient → Recipient

Another example: Patient reported severe headache.

- Patient → Experiencer
- Headache → Symptom

Thus, syntactic parsing helps the system assign appropriate semantic roles.

2. Determine whether the assigned semantic roles are appropriate.

Ans:

Entity	Given Role	Evaluation
Doctor	Agent	Appropriate
Medicine	Instrument	Partially inappropriate
Patient	Recipient	Appropriate in first sentence
Headache	Symptom	Appropriate

The main issue is Medicine → Instrument.

In "Doctor prescribed medicine to patient", medicine is better represented as the Theme/Object of the prescribing action rather than an instrument.

3. Evaluate potential errors that could arise from incorrect parsing.

Ans:

Incorrect parsing can cause:

- Wrong patient information.
- Incorrect medication relationships.
- Incorrect diagnosis interpretation.
- Wrong treatment information.
- Incorrect medical records.
- Unsafe clinical decisions.

For example, confusing the patient and doctor roles could result in incorrect interpretation of who received or prescribed treatment.

4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Ans:

Healthcare NLP systems should use:

- Medical-specific parsers.
- Dependency parsing.

- Semantic role labelling.
- Medical Named Entity Recognition.
- Context-aware language models.
- Medical dictionaries and ontologies.
- Human validation for critical information.

Conclusion: Syntax-driven semantic analysis converts sentence structure into meaningful medical relationships. Accurate parsing and semantic role assignment are essential for reliable healthcare NLP systems.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

User Query	Possible Sense 1	Possible Sense 2
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